



W1-2-60-1-6

JOMO KENYATTA UNIVERSITY OF AGRICULTURE AND TECHNOLOGY

UNIVERSITY EXAMINATIONS 2018/2019

**FIRST SEMESTER EXAMINATION FOR THE DEGREE OF BACHELOR OF
SCIENCE IN BIOSTATISTICS, OPERATIONS RESEARCH, STATISTICS,
ACTUARIAL SCIENCE AND FINANCIAL ENGINEERING.**

STA 2104: CALCULUS FOR STATISTICS I

DATE: DECEMBER 2018

TIME: 2HOURS

INSTRUCTIONS: Answer question one (compulsory) any other two questions

QUESTION ONE (30 MARKS)

a) Evaluate the following limits

(i) $\lim_{x \rightarrow -1} \left(\frac{x+1}{x^2 - x - 2} \right)$ (2marks)

(ii) $\lim_{x \rightarrow \infty} \left(\frac{16 - 6x - 5x^3}{2x^2 + 14x + 5} \right)$ (3marks)

(iii) $\lim_{x \rightarrow -4} \left(\frac{\frac{1}{\sqrt{13+x}} - \frac{1}{3}}{x+4} \right)$ (3marks)

b) The sales (in million Dollars) of a laser disc recording of a hit movie, t years from the date of release is given by $S(t) = 100 \left(\frac{t^2 - 4t + 4}{t^2 + 4} \right), 0 \leq t \leq \infty$

(i) Find the rate at which the sales are changing at time t .

(ii) How fast are the sales changing at the time the laser discs are released ($t = 0$)? Two years from the date of release. (4 marks)

c) Find the value of x for which the function $f(x) = \begin{cases} \frac{x^2 - 16}{x - 4} & x \neq 4 \\ 3 & x = 4 \end{cases}$ is continuous. (3 marks)

d) Use the first principle of differentiation to find $\frac{dy}{dx}$ given $f(x) = \frac{x}{1-7x}$ (4 marks)

- e) State the Rolle's Theorem hence use it to determine whether the function $f(x) = 5x^{2/3} - x^{5/3}$ on $[0, 5]$ satisfies the hypothesis of Rolle's theorem on the indicated interval and find the values of x in the interval $[0, 5]$ that satisfies the conclusion of the theorem (4 marks)
- f) A certain species of turtles faces extinction since dealers called truckloads of turtle's eggs to be sold as aphrodisiacs. After severe conservation measures were implemented, it is hoped that the turtle population will grow according to the rule $N(t) = 2t^3 + 3t^2 - 4t + 1000$, $0 \leq t \leq 10$ where $N(t)$ denotes the population at the end of the year t . Find the rate of growth of the turtle population when $t = 2$ and $t = 8$. What will be the population be 10 years after the conservation measures are implemented. (4 marks)
- g) Use linear approximation to estimate the value of $\cos(32^\circ)$ (3 marks)

QUESTION TWO (20 MARKS)

- a) Find the values of the constant a and b for which the given function is continuous everywhere in the interval $(-\infty, \infty)$. $f(x) = \begin{cases} -5 & \text{if } x < -1 \\ ax + b & \text{if } -1 \leq x < 2 \\ -10x & \text{if } x > 2 \end{cases}$ (6 marks)
- b) A major audio-tape manufacturer is willing to make x thousand 10 packs of metal alloy audiocassette tapes per week available in the market place when the wholesale price is \$ p per 10-pack. It is known that the relationship between x and p is governed by the supply equation $x^2 - 3xp + p^2 = 5$. How fast is the supply of tapes changing when the price per 10-pack is \$11, the quantity supplied is 4,000 10-packs and the wholesale price per 10-pack is increasing at the rate of 10 percent per 10-pack per week (4 marks)
- c) Find the derivative of the following functions.
- (i) $y(x) = \left(\frac{3x+1}{1-2x^2} \right)^6 + \frac{40}{\sqrt[3]{x^3 - 6x^2 + 1}}$ (5 marks)
- (ii) $y(x) = a^{x^2} + e^{2x^2 - 4x} + \ln(\sqrt{x^2 + 4x})$ (5 marks)

QUESTION THREE (20 MARKS)

- a) Identify and classify the stationary points of the function $f(x) = 3x^3 - 36x^2 + 135x - 13$ (9 marks)
- b) For a firm under perfect competition, the total cost function in shillings is given by $C(x) = \frac{x^3}{25} - \frac{9x^2}{10} + 10x + 12$. If the price per unit is sh.4, will the firm continue production? (6 marks)
- c) Find $\frac{dy}{dx}$ given $y(x) = (x^3 + \sqrt{x})^3 \sin^{-1}(4x) + x^x$ (5 marks)

QUESTION FOUR (20 MARKS)

a) Find $\frac{dy}{dx}$ given

(i) $xy = 6e^{2x-3y}$ at $(3, 2)$ (3 marks)

(ii) $\cos(x + y) = \sin x \sin y$ (3 marks)

b) Use linear approximation to estimate $\sqrt[3]{62}$ (4 marks)

c) State the mean value theorem hence or otherwise, show that the function $f(x) = 1 + |x - 1|$ on $[0, 3]$ neither satisfies the hypothesis nor the conclusion of the mean value theorem on the indicated interval. (4 marks)

d) Find the equation of the tangent line and normal line to the curve $12(x^2 + y^2) = 25xy$ at $(3, 4)$ (6 marks)